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<div class="logo-container">

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```
<div class="center-title">Dokumentasi Teknis Lengkap & API Reference</div>
<div class="center-subtitle">(Dokumentasi Arsitektur Kode Camelio Finance)</div>
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<div class="intro-card">
<p>Halo! Selamat datang di <b>Dokumentasi Source Code Camelio Finance</b>.</p>
<p>Dokumen ini dirancang khusus untuk memberikan gambaran arsitektur sistem (High-Level) sekaligus
bedah kode spesifik (Low-Level) pada keseluruhan project ini.</p>
<p>Dokumen ini mengekstrak secara otomatis semua file, class, dan fungsi dari dalam <i>source code</i>
aplikasi untuk memastikan tidak ada file yang terlewat. Mari kita mulai!</p>
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<h2 class="section-title"> 1. Tumpukan Teknologi (Tech Stack) Utama</h2>
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- **Framework:** Flutter (Dart)
- **Architecture:** MVVM dengan Provider & GetIt (Service Locator)
- **Backend & Auth:** Firebase Auth, Cloud Firestore
- **Kecerdasan Buatan (AI):** Google Gemini 1.5 Flash (Vision & LLM)
- **Cloud Storage:** Cloudinary (Image Optimization & Hosting)

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<h2 class="section-title"> 2. Struktur Database (Firestore NoSQL)</h2>
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1. **users:** Profil Pengguna (balance, monthlyBudget, dll)
2. **wallets:** Data E-Wallet dan Rekening Bank
3. **transactions:** History Pemasukan/Pengeluaran beserta detail item (OCR AI)
4. **notifications & rating:** Log peringatan AI Insight dan Feedback User.

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<h2 class="section-title"> 3. Alur Kerja Integrasi AI (Gemini 1.5)</h2>
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- **Scan Struk OCR:** Foto diupload ke Cloudinary -> URL dikirim ke Gemini Vision -> Ekstrak Entity JSON -> Simpan ke Firestore.
- **Input Suara/Teks Pintar:** Prompt Chat dikirim ke Gemini LLM -> Ekstrak nominal dan kategori secara otomatis.
- **AI Financial Advisor:** Sistem fetch data transaksi 30 hari -> Dikirim ke Gemini -> Gemini berperan sebagai Konsultan Bisnis UMKM dan mengembalikan rekomendasi.

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<h2 class="section-title"> 4. Bedah Kode Sumber (API Reference)</h2>
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Berikut adalah rincian lengkap dari seluruh file .dart yang ada di folder lib/, beserta class dan fungsi yang ada di dalamnya:

lib/app.dart

- **Panjang Baris:** 53 lines
- **Class Name:** `SmartFinanceApp`
- **Method / Fungsi:** `build()`

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lib/core/constants/api_keys.dart

- **Panjang Baris:** 78 lines
- **Class Name:** `ApiKeys`
- **Method / Fungsi:** `initRemoteConfig()`

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lib/core/constants/app_colors.dart

- **Panjang Baris:** 42 lines
- **Class Name:** `AppColors`

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lib/core/constants/app_constants.dart

- **Panjang Baris:** 6 lines
- **Class Name:** `AppConstants`

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lib/core/locator.dart

- **Panjang Baris:** 24 lines
- **Method / Fungsi:** `setupLocator()`

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lib/core/theme/app_theme.dart

- **Panjang Baris:** 107 lines
- **Class Name:** `AppTheme`

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lib/core/utils/brand_utils.dart

- **Panjang Baris:** 88 lines
- **Class Name:** `BrandUtils`
- **Method / Fungsi:** `getBrandIcon()`

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lib/core/utils/currency_formatter.dart

- **Panjang Baris:** 31 lines
- **Class Name:** `CurrencyFormatter`
- **Method / Fungsi:** `format()`

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lib/core/utils/date_formatter.dart

- **Panjang Baris:** 19 lines
- **Class Name:** `DateFormatter`
- **Method / Fungsi:** `formatMedium()`, `formatTime()`

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lib/data/models/e_wallet_model.dart

- **Panjang Baris:** 82 lines
- **Class Name:** `EWalletModel`
- **Method / Fungsi:** `toMap()`

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lib/data/models/expense_model.dart

- **Panjang Baris:** 105 lines
- **Class Name:** `ExpenseItem`, `ExpenseModel`
- **Method / Fungsi:** `toJson()`, `toFirestore()`

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lib/data/models/notification_model.dart

- **Panjang Baris:** 50 lines
- **Class Name:** `NotificationModel`
- **Method / Fungsi:** `toMap()`

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lib/data/models/transaction_model.dart

- **Panjang Baris:** 70 lines
- **Class Name:** `TransactionModel`
- **Method / Fungsi:** `toMap()`

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lib/data/models/user_model.dart

- **Panjang Baris:** 73 lines
- **Class Name:** `UserModel`
- **Method / Fungsi:** `toFirestore()`

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lib/data/services/auth_service.dart

- **Panjang Baris:** 83 lines
- **Class Name:** `AuthService`
- **Method / Fungsi:** `signInAnonymously()`, `signInWithGoogle()`, `registerWithEmail()`, `signInWithEmail()`, `signOut()`

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lib/data/services/cloudinary_service.dart

- **Panjang Baris:** 64 lines
- **Class Name:** `CloudinaryService`

- **Method / Fungsi:** `<code>uploadImage()</code>`

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lib/data/services/csv_logger_service.dart

- **Panjang Baris:** 55 lines
- **Class Name:** `<code>CsvLoggerService</code>`
- **Method / Fungsi:** `<code>logReceiptScan()</code>`

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lib/data/services/firestore_service.dart

- **Panjang Baris:** 344 lines
- **Class Name:** `<code>FirestoreService</code>`
- **Method / Fungsi:** `<code>createUserProfile()</code>`, `<code>updateUserProfile()</code>`, `<code>saveEWallet()</code>`, `<code>deleteEWallet()</code>`, `<code>updateEWalletBalance()</code>`, `<code>updateEWalletAccountNumber()</code>`, `<code>addTransaction()</code>`, `<code>deleteTransaction()</code>`, `<code>getTransactionById()</code>`, `<code>submitRating()</code>`, `<code>addNotification()</code>`, `<code>markNotificationAsRead()</code>`, `<code>deleteNotification()</code>`, `<code>deleteAllNotifications()</code>`

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lib/data/services/gemini_ai_service.dart

- **Panjang Baris:** 214 lines
- **Class Name:** `<code>GeminiAIService</code>`
- **Method / Fungsi:** `<code>classifyTransaction()</code>`, `<code>classifyReceiptFromImage()</code>`, `<code>getFinancialAdvice()</code>`, `<code>getPortfolioInsight()</code>`, `<code>generateContent()</code>`

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lib/data/services/gmail_service.dart

- **Panjang Baris:** 128 lines
- **Class Name:** `<code>GmailService</code>`
- **Method / Fungsi:** `<code>isSignedIn()</code>`, `<code>signIn()</code>`, `<code>signOut()</code>`, `<code>getGmailApi()</code>`, `<code>sendConfirmationEmail()</code>`

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lib/data/services/notification_service.dart

- **Panjang Baris:** 70 lines
- **Class Name:** `<code>NotificationService</code>`
- **Method / Fungsi:** `<code>init()</code>`, `<code>showNotification()</code>`

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lib/firebase_options.dart

- **Panjang Baris:** 95 lines
- **Class Name:** `DefaultFirebaseOptions`

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lib/main.dart

- **Panjang Baris:** 47 lines
- **Method / Fungsi:** `main()`

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lib/providers/ai_insight_provider.dart

- **Panjang Baris:** 48 lines
- **Class Name:** `AllInsightProvider`
- **Method / Fungsi:** `askForFinancialInsight()`, `clearConversation()`

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lib/providers/auth_provider.dart

- **Panjang Baris:** 301 lines
- **Class Name:** `AuthProvider`
- **Method / Fungsi:** `loadUserProfile()`, `updateProfile()`, `updateProfilePicture()`, `updateBudget()`, `deleteProfilePicture()`, `signInWithGoogle()`, `signInWithEmail()`, `registerWithEmail()`, `signOut()`, `_handleFirebaseError()`, `_setLoading()`, `_setError()`, `_clearError()`, `clearError()`, `dispose()`

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lib/providers/e_wallet_provider.dart

- **Panjang Baris:** 349 lines
- **Class Name:** `EWalletProvider`
- **Method / Fungsi:** `loadWallets()`, `saveWallet()`, `deleteWallet()`, `updateWalletBalance()`, `updateWalletAccountNumber()`, `syncWithGmail()`, `_parseGeminiResponse()`, `dispose()`

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lib/providers/notification_provider.dart

- **Panjang Baris:** 115 lines
- **Class Name:** `NotificationProvider`
- **Method / Fungsi:** `loadNotifications()`, `addNotification()`, `sendTestNotification()`, `Function()`, `markAsRead()`, `deleteNotification()`, `deleteAllNotifications()`, `dispose()`

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lib/providers/settings_provider.dart

- **Panjang Baris:** 298 lines
- **Class Name:** `SettingsProvider`
- **Method / Fungsi:** `toggleTheme()`, `toggleNotifications()`, `toggleGmailSync()`, `resetUserSpecificSettings()`, `setLanguage()`, `setAccountType()`, `toggleCurrency()`, `_checkAllPermissions()`, `requestPermission()`, `requestStoragePermission()`, `_fetchRealtimeExchangeRate()`, `_loadSettings()`, `translate()`, `translateCategory()`

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lib/providers/transaction_provider.dart

- **Panjang Baris:** 131 lines
- **Class Name:** `TransactionProvider`
- **Method / Fungsi:** `toggleBalanceVisibility()`, `loadTransactions()`, `addTransaction()`, `deleteTransaction()`, `clearTransactions()`, `dispose()`

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lib/ui/pages/ai_advisor/ai_insight_page.dart

- **Panjang Baris:** 423 lines
- **Class Name:** `AIInsightPage`, `_AIInsightPageState`
- **Method / Fungsi:** `initState()`, `_generateInsight()`, `build()`

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lib/ui/pages/auth/login_page.dart

- **Panjang Baris:** 281 lines
- **Class Name:** `LoginPage`, `_LoginPageState`
- **Method / Fungsi:** `_showLoginSuccess()`, `_handleLogin()`, `build()`

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lib/ui/pages/auth/register_page.dart

- **Panjang Baris:** 228 lines
- **Class Name:** `RegisterPage`, `_RegisterPageState`
- **Method / Fungsi:** `_handleRegister()`, `build()`

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lib/ui/pages/budget/budget_page.dart

- **Panjang Baris:** 266 lines
- **Class Name:** `BudgetPage`, `_BudgetPageState`
- **Method / Fungsi:** `initState()`, `_formatNumber()`, `dispose()`

</code>, <code>build()</code>

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lib/ui/pages/dashboard/dashboard_page.dart

- **Panjang Baris:** 245 lines
- **Class Name:** <code>DashboardPage</code>, <code>_DashboardPageState</code>
- **Method / Fungsi:** <code>build()</code>, <code>_showZoomableImage()</code>

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lib/ui/pages/dashboard/widgets/action_button_row.dart

- **Panjang Baris:** 147 lines
- **Class Name:** <code>ActionButtonRow</code>
- **Method / Fungsi:** <code>build()</code>, <code>_buildSummaryCard()</code>

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lib/ui/pages/dashboard/widgets/balance_card.dart

- **Panjang Baris:** 179 lines
- **Class Name:** <code>BalanceCard</code>
- **Method / Fungsi:** <code>build()</code>

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lib/ui/pages/dashboard/widgets/feature_cards.dart

- **Panjang Baris:** 153 lines
- **Class Name:** <code>FeatureCards</code>
- **Method / Fungsi:** <code>build()</code>, <code>_buildCard()</code>

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lib/ui/pages/dashboard/widgets/recent_transactions.dart

- **Panjang Baris:** 232 lines
- **Class Name:** <code>RecentTransactions</code>
- **Method / Fungsi:** <code>build()</code>

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lib/ui/pages/e_wallet/add_wallet_page.dart

- **Panjang Baris:** 625 lines
- **Class Name:** <code>AddWalletPage</code>, <code>_AddWalletPageState</code>
- **Method / Fungsi:** <code>initState()</code>, <code>_onNameChanged()</code>, <code>dispose()</code>, <code>build()</code>

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lib/ui/pages/e_wallet/e_wallet_detail_page.dart

- **Panjang Baris:** 868 lines
- **Class Name:** `EWalletDetailPage`, `_EWalletDetailPageState`
- **Method / Fungsi:** `build()`, `_buildBalanceCard()`, `_getWalletBranding()`, `_buildTransactionTabs()`, `_showEditBalanceDialog()`, `_showEditAccountDialog()`, `_showDeleteConfirmDialog()`

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lib/ui/pages/e_wallet/e_wallet_page.dart

- **Panjang Baris:** 481 lines
- **Class Name:** `EWalletPage`, `_EWalletPageState`
- **Method / Fungsi:** `initState()`, `dispose()`, `build()`, `_buildAddWalletCard()`, `_buildWalletItem()`

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lib/ui/pages/main/main_page.dart

- **Panjang Baris:** 2406 lines
- **Class Name:** `MainPage`, `_MainPageState`
- **Method / Fungsi:** `initState()`, `build()`, `_buildNavItem()`, `_showLogoutDialog()`, `_showRatingDialog()`, `_buildProfileDrawer()`, `_buildDrawerItem()`, `_buildDrawerSectionTitle()`, `_buildDrawerBox()`, `_showEditProfileModal()`, `_showImageSourcePicker()`, `_buildSourceOption()`, `_viewProfilePhoto()`, `_pickAndUploadImage()`, `_showConfirmDeletePhoto()`, `_showAddMenu()`, `_buildMenuOption()`, `_showAllBottomSheet()`, `_handleExportPdf()`, `_showPopup()`, `_containsKeyword()`, `_extractCategoryLocally()`

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lib/ui/pages/notification/notification_page.dart

- **Panjang Baris:** 415 lines
- **Class Name:** `NotificationPage`, `_NotificationPageState`
- **Method / Fungsi:** `initState()`, `dispose()`, `build()`, `_buildEmptyState()`, `_buildNotificationItem()`, `_buildNotificationIcon()`, `_showDeleteAllDialog()`

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lib/ui/pages/portfolio/portfolio_page.dart

- **Panjang Baris:** 1066 lines
- **Class Name:** `PortfolioPage`, `_PortfolioPageState`
- **Method / Fungsi:** `build()`, `_getChartTitle()`, `_buildTabItem()`, `_buildSummaryCards()`, `_buildCard()`, `_buildChart()`, `_buildCalendarStrip()`, `_buildChartContent()`, `_buildWeeklyChart()`, `_buildMonthlyChart()`


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<code>_buildYearlyChart()</code>, <code>_buildChartContainer()</code>, <code>Function()</code>, <code>_buildPieChart()</code>, <code>_buildWalletPieChart()</code>, <code>_showAllInsight()</code>
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lib/ui/pages/settings/about_camelio_page.dart

- **Panjang Baris:** 271 lines
- **Class Name:** <code>AboutCamelioModal</code>, <code>_AboutCamelioModalState</code>
- **Method / Fungsi:** <code>build()</code>, <code>_buildAboutFeatureItem()</code>

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lib/ui/pages/settings/help_page.dart

- **Panjang Baris:** 161 lines
- **Class Name:** <code>HelpFAQPage</code>, <code>_HelpFAQPageState</code>
- **Method / Fungsi:** <code>build()</code>, <code>_buildFAQItem()</code>

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lib/ui/pages/settings/pdf_generator.dart

- **Panjang Baris:** 552 lines
- **Class Name:** <code>PdfGenerator</code>
- **Method / Fungsi:** <code>generateAndShare()</code>, <code>_generateAIAdvice()</code>, <code>_buildSummaryBox()</code>, <code>buildExportChart()</code>

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lib/ui/pages/settings/permissions_page.dart

- **Panjang Baris:** 235 lines
- **Class Name:** <code>PermissionsPage</code>
- **Method / Fungsi:** <code>build()</code>, <code>_buildPermissionTile()</code>

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lib/ui/pages/settings/policy_page.dart

- **Panjang Baris:** 159 lines
- **Class Name:** <code>PrivacyPolicyPage</code>, <code>_PrivacyPolicyPageState</code>
- **Method / Fungsi:** <code>build()</code>

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lib/ui/pages/settings/privacy_page.dart

- **Panjang Baris:** 157 lines
- **Class Name:** <code>PrivacyControlsPage</code>, <code>_PrivacyControlsPageState</code>
- **Method / Fungsi:** <code>build()</code>, <code>_buildInfoCard()</code>

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lib/ui/pages/settings/report_generator.dart

- **Panjang Baris:** 150 lines
- **Class Name:** `ReportGenerator`
- **Method / Fungsi:** `buildDashboardReport()`, `buildEWalletReport()`

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lib/ui/pages/settings/settings_page.dart

- **Panjang Baris:** 287 lines
- **Class Name:** `SettingsPage`
- **Method / Fungsi:** `build()`, `_buildSectionTitle()`, `_buildSettingCard()`, `_buildSwitchTile()`, `_buildNavigationTile()`

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lib/ui/pages/settings/version_page.dart

- **Panjang Baris:** 346 lines
- **Class Name:** `VersionHistoryPage`, `_VersionHistoryPageState`
- **Method / Fungsi:** `build()`, `_buildVersionItem()`

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lib/ui/pages/splash/splash_page.dart

- **Panjang Baris:** 122 lines
- **Class Name:** `SplashPage`, `_SplashPageState`
- **Method / Fungsi:** `initState()`, `build()`

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lib/ui/pages/transaction/add_transaction_page.dart

- **Panjang Baris:** 482 lines
- **Class Name:** `CurrencyInputFormatter`, `AddTransactionPage`, `_AddTransactionPageState`
- **Method / Fungsi:** `dispose()`, `build()`

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lib/ui/pages/transaction/scan_page.dart

- **Panjang Baris:** 14 lines
- **Class Name:** `ScanPage`
- **Method / Fungsi:** `build()`

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lib/ui/pages/transaction/scan_receipt_page.dart

- **Panjang Baris:** 868 lines

- **Class Name:** `ScanReceiptPage`, `_ScanReceiptPageState`
- **Method / Fungsi:** `_processReceipt()`, `_saveTransaction()`, `_confirmDeleteItem()`, `_showEditTotalDialog()`, `_editItemPrice()`, `build()`

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lib/ui/pages/transaction/top_up_page.dart

- **Panjang Baris:** 14 lines
- **Class Name:** `TopUpPage`
- **Method / Fungsi:** `build()`

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lib/ui/pages/transaction/transaction_list_page.dart

- **Panjang Baris:** 186 lines
- **Class Name:** `TransactionListPage`, `_TransactionListPageState`
- **Method / Fungsi:** `build()`

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lib/ui/pages/transaction/transfer_page.dart

- **Panjang Baris:** 16 lines
- **Class Name:** `TransferPage`
- **Method / Fungsi:** `build()`

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lib/ui/widgets/custom_popup.dart

- **Panjang Baris:** 77 lines
- **Class Name:** `CustomPopup`
- **Method / Fungsi:** `show()`

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lib/ui/widgets/custom_text_field.dart

- **Panjang Baris:** 101 lines
- **Class Name:** `CustomTextField`, `_CustomTextFieldState`
- **Method / Fungsi:** `initState()`, `build()`

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lib/ui/widgets/transaction_card.dart

- **Panjang Baris:** 367 lines
- **Class Name:** `TransactionCardListTile`
- **Method / Fungsi:** `build()`, `_showItemsBottomSheet()`, `_buildTransactionIcon()`

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